



Effectiveness of natural language intelligence technology in chronic diseases nursing: A systematic review and meta-analysis

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ABSTRACT

Background: Chronic diseases pose a heavy global burden, with challenges in utilizing unstructured data for continuous care. Natural language intelligence technology (NLIT) shows potential in addressing these issues.

Methods: This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, with its protocol registered on PROSPERO (CRD420251006387). We searched PubMed, Web of Science, Embase, and Scopus databases for studies published between January 2014 and January 2024. The risk of bias in included studies was assessed using the Newcastle-Ottawa Scale (NOS) for cohort studies and the Quality Assessment of Diagnostic Accuracy Studies-Comparative (QUADAS-C) scale for diagnostic accuracy studies.

Results: A total of 6 studies involving 28,323 participants were included. NLIT showed beneficial effectiveness: in chronic obstructive pulmonary disease (COPD) management, NLIT was delivered via an Artificial Neural Network (ANN)-powered smartphone app (Re-Admit) that processed patient symptom reports and EHR data, reducing 30-day readmission rates by 27.9%; in diabetes care, it improved self-management behaviors ($P < 0.05$) and diabetic retinopathy referral compliance by 19.34% ($P < 0.01$); in stroke rehabilitation, it enhanced motor function (Wolf test, $P = 0.02$) and shoulder range of motion ($P \leq 0.01$). Technical performance included 88.46% accuracy in predicting COPD exacerbations and 92% diagnostic accuracy for diabetes using Generative Pre-trained Transformer 4 (GPT-4). Meta-analysis of binary compliance data from two studies (Musculoskeletal [MSK] conditions and diabetes care) showed that NLIT demonstrated measurable benefits for clinical outcomes, with a pooled relative risk (RR) of 1.20 (95% confidence interval [CI]: 1.03–1.40, $P = 0.02$); however, substantial heterogeneity was observed ($I^2 = 75\%$, $P = 0.05$). Five studies were rated as low risk of bias, with one having moderate risk due to hypothetical data.

Conclusions: NLIT is valuable for personalized care and resource optimization but faces challenges like data heterogeneity and bias, requiring further refinement.

What is already known

- Chronic diseases impose a heavy global burden, requiring continuous monitoring and long-term care, which places immense pressure on healthcare systems - a challenge compounded by difficulties in

extracting valuable information from unstructured data sources such as electronic health records and clinical notes.

- Natural language intelligence technology has shown potential in addressing these challenges by improving diagnostic accuracy and facilitating personalized intervention.

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